CS 1302

Computer Networks

— Unit - 5 —

— Application Layer —

- **Text Book**

  Behrouz .A. Forouzan, “Data communication and Networking”, Tata McGrawHill, 2004

Domain Name System

25.1 Name Space

Flat Name Space

Hierarchical Name Space

25.2 Domain Name Space

Label

Domain Name

Figure 25.1 Domain name space

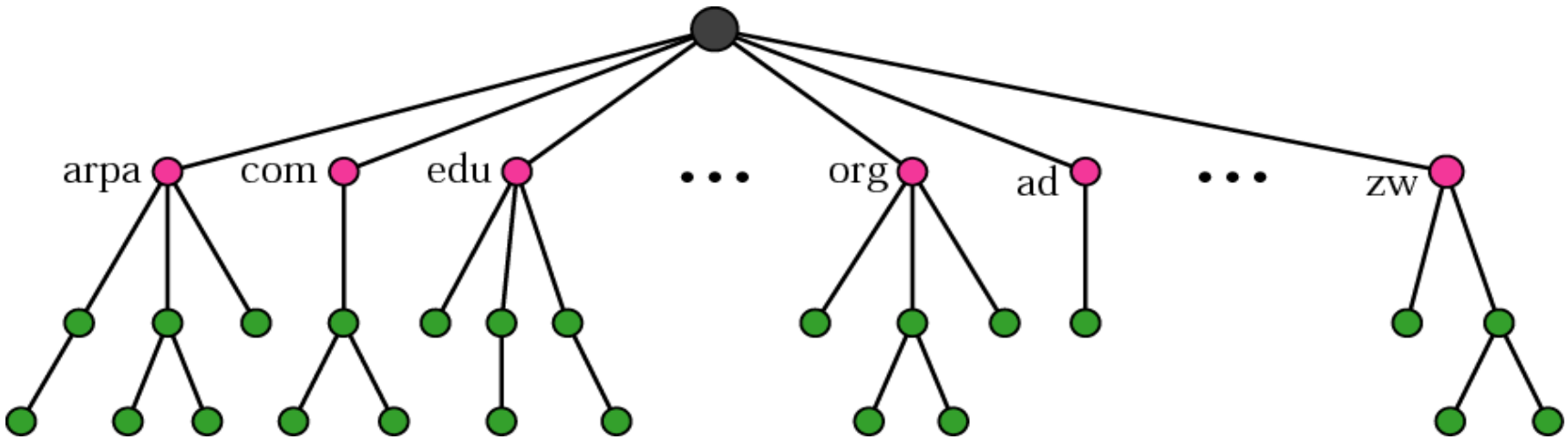


Figure 25.2 Domain names and labels

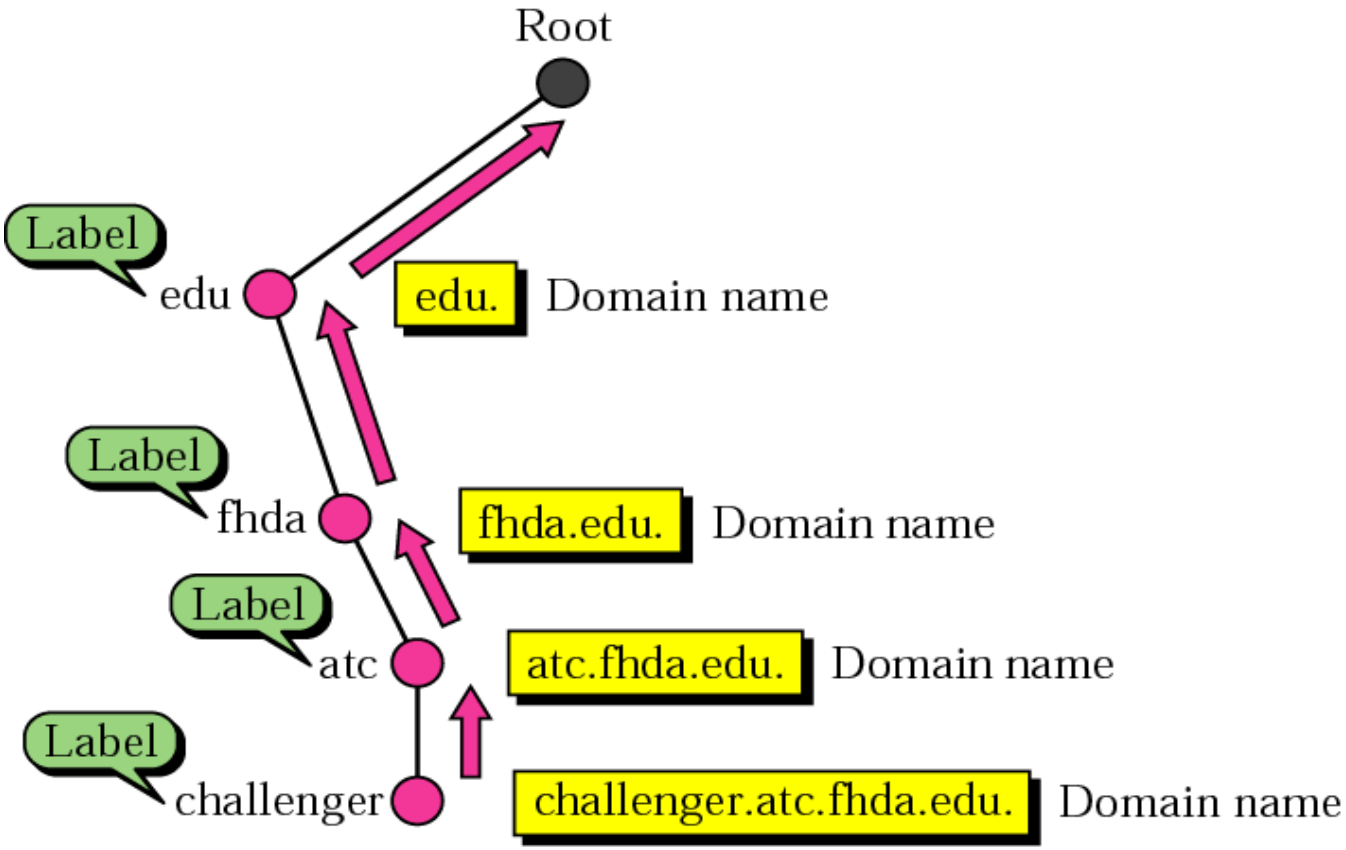
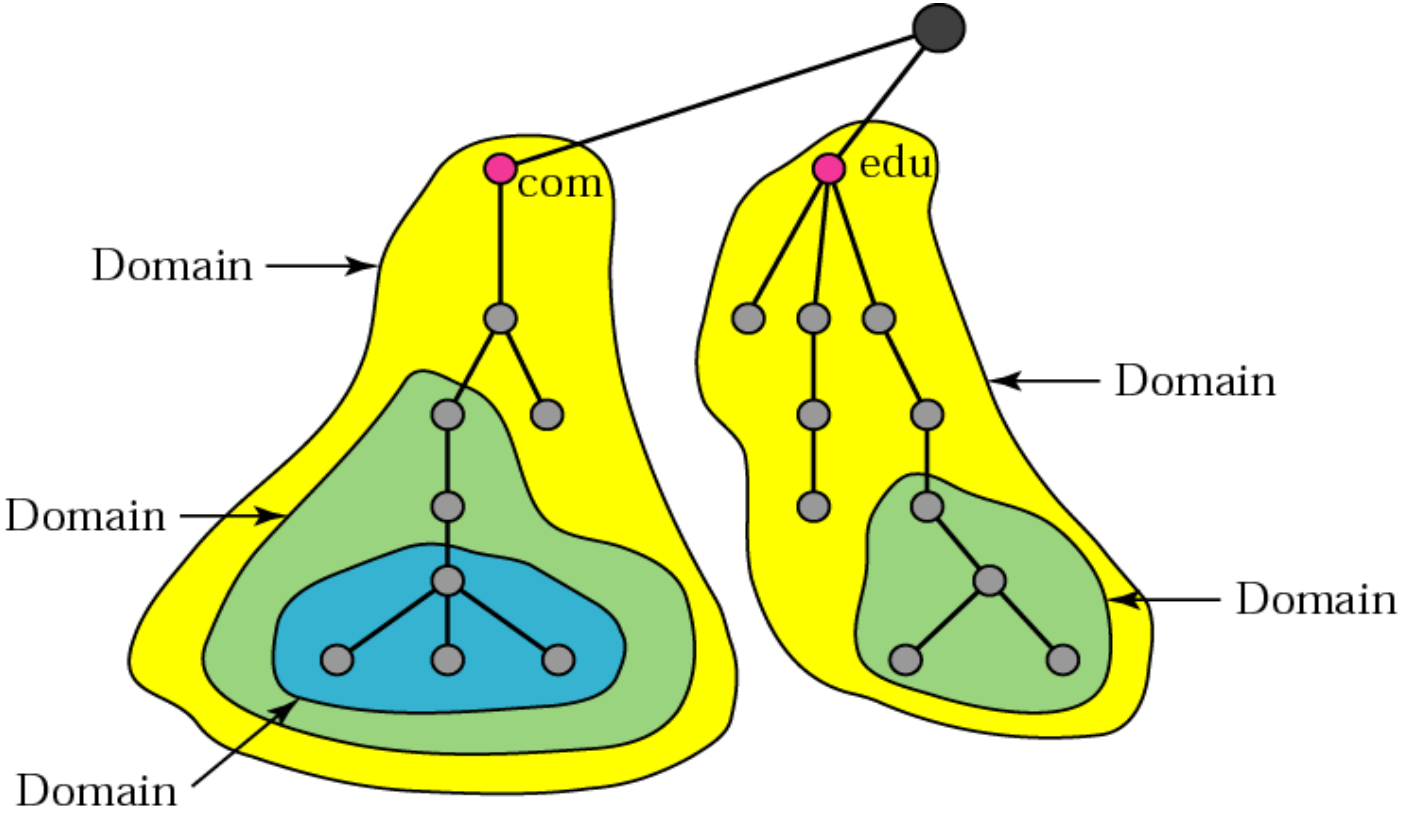


Figure 25.3 FQDN and PQDN

FQDN	PQDN
challenger.atc.fhda.edu.	challenger.atc.fhda.edu
cs.hmme.com.	cs.hmme
www.funny.int.	www

Figure 25.4 Domains



25.3 Distribution of Name Spaces

Hierarchy of Name Servers

Zone What a server is responsible for or has authority

Root Server Server that consists of whole tree. Only has references. Root servers are distributed all over the world

Primary and Secondary Servers

Primary- Stores a file about zone for which it is having authority
Secondary – Transfers complete information from another server

Figure 25.5 Hierarchy of name servers

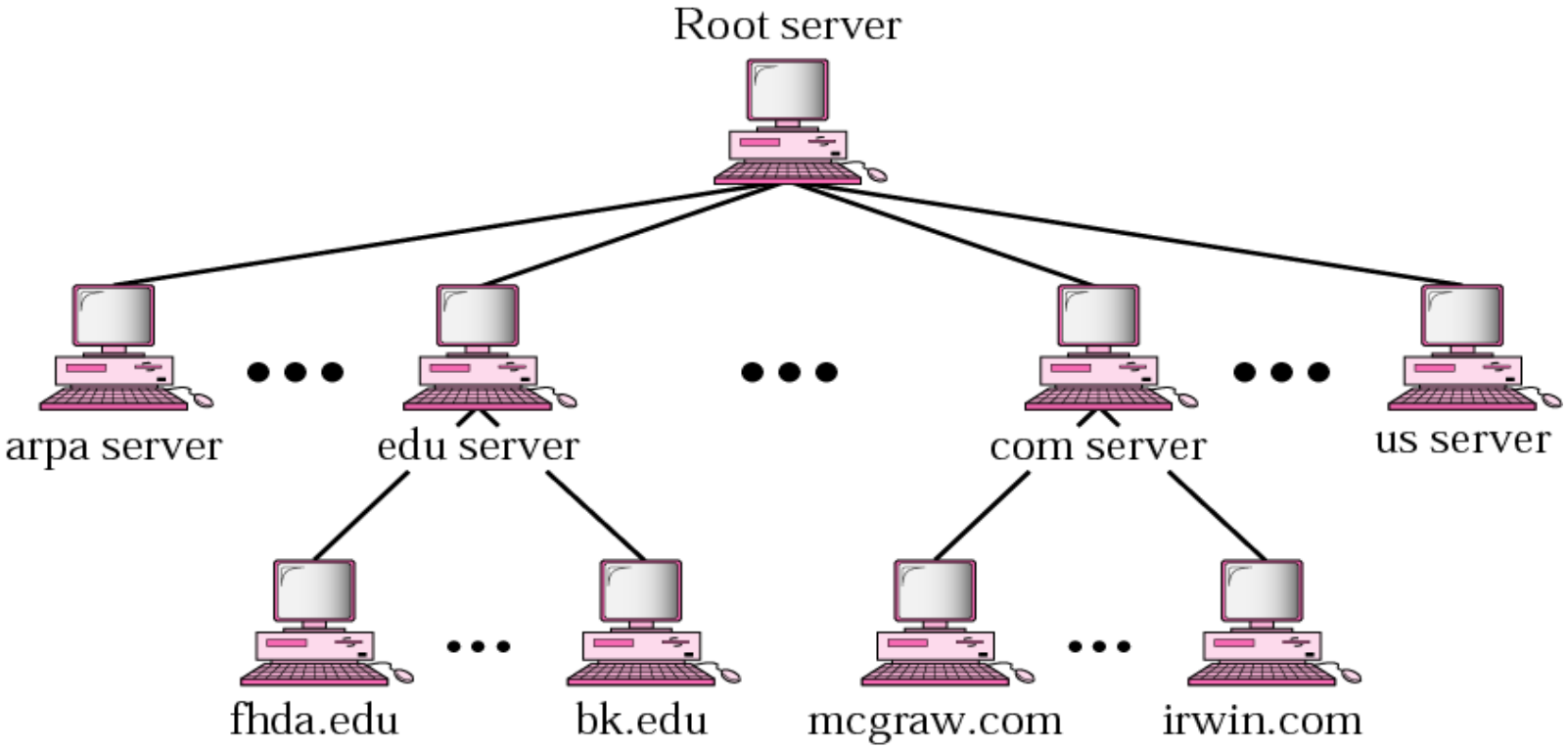
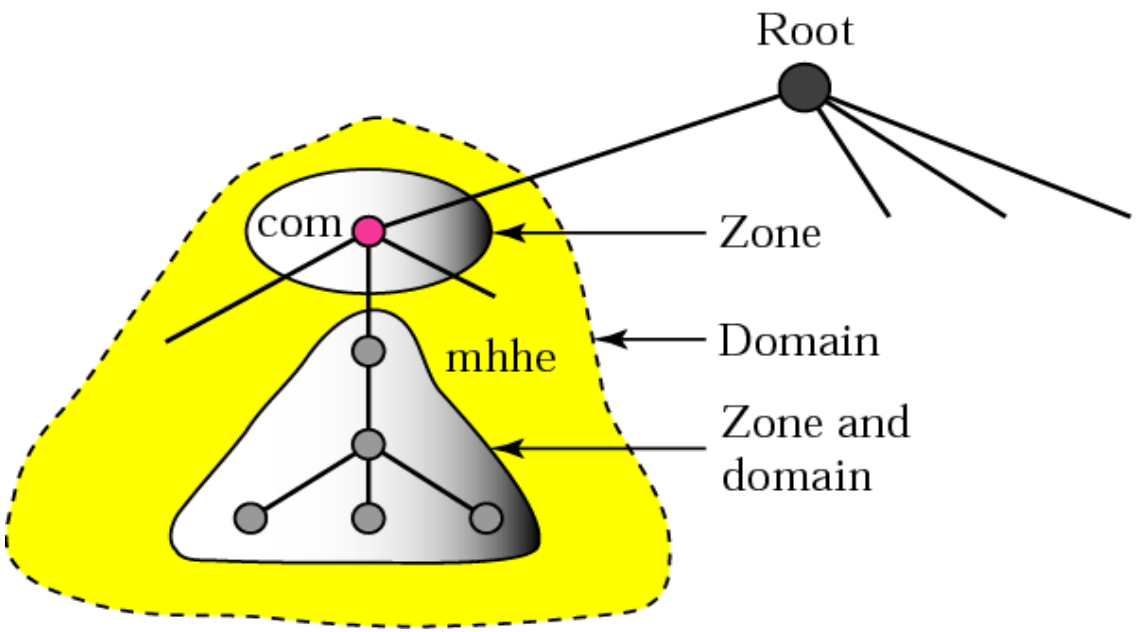


Figure 25.6 Zones and domains



A primary server loads all information from the disk file; the secondary server loads all information from the primary server.

25.4 DNS In The Internet

Generic Domain

Registered hosts according to their generic behavior

Country Domain

Organizational or specific national designations

Inverse Domain

Figure 25.7 DNS in the Internet

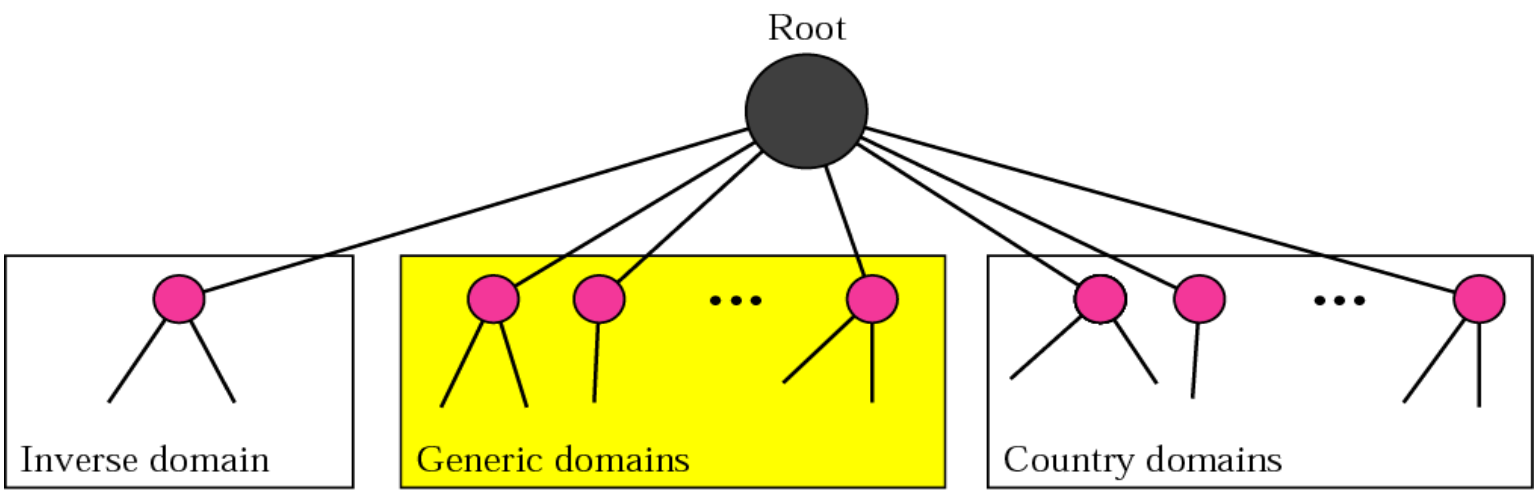


Figure 25.8 Generic domains

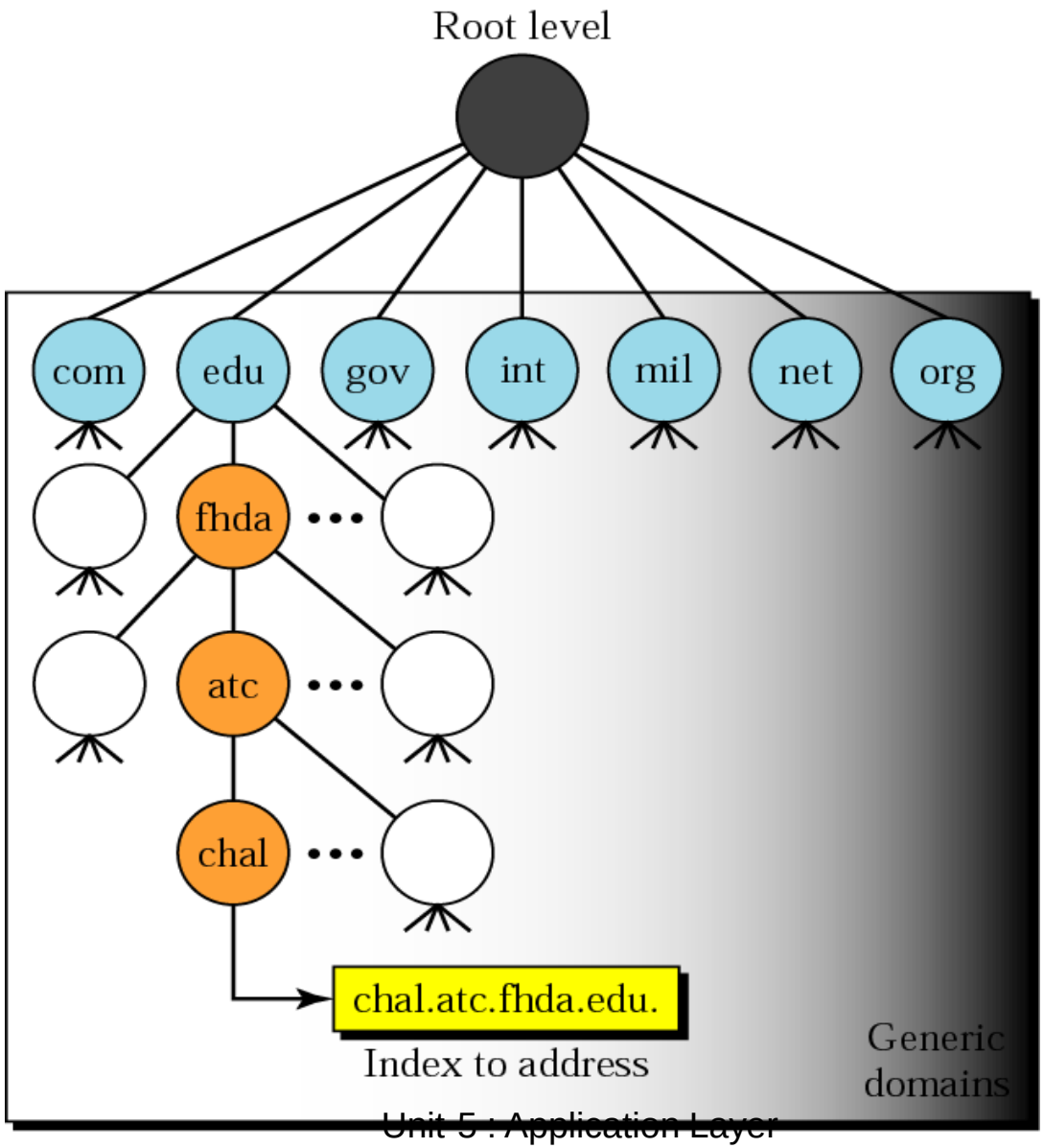


Table 25.1 Generic domain labels

Label	Description
com	Commercial organizations
edu	Educational institutions
gov	Government institutions
int	International organizations
mil	Military groups
net	Network support centers
org	Nonprofit organizations

Table 25.2 New generic domain labels

Label	Description
aero	Airlines and aerospace companies
biz	Businesses or firms (similar to com)
coop	Cooperative business organizations
info	Information service providers
museum	Museums and other nonprofit organizations
name	Personal names (individuals)
pro	Professional individual organizations

Figure 25.9 Country domains

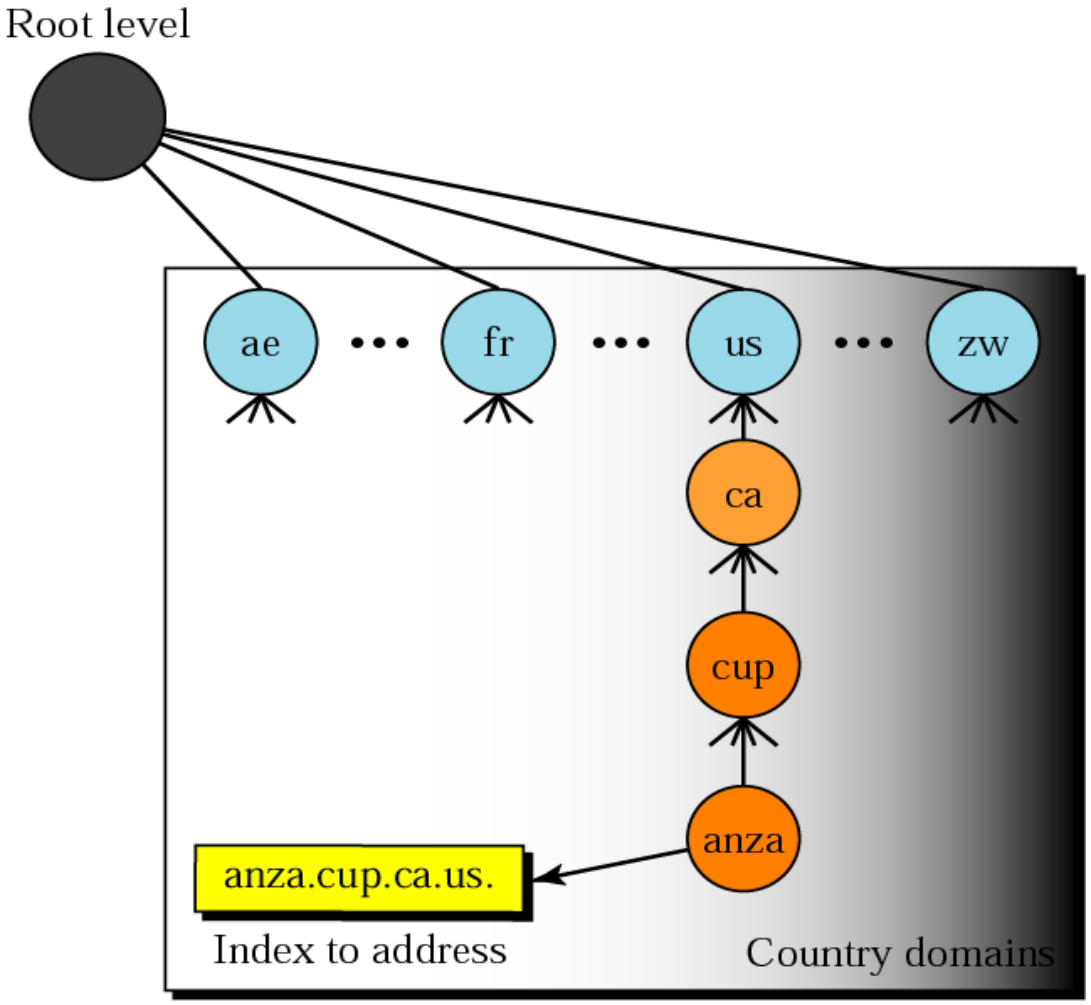
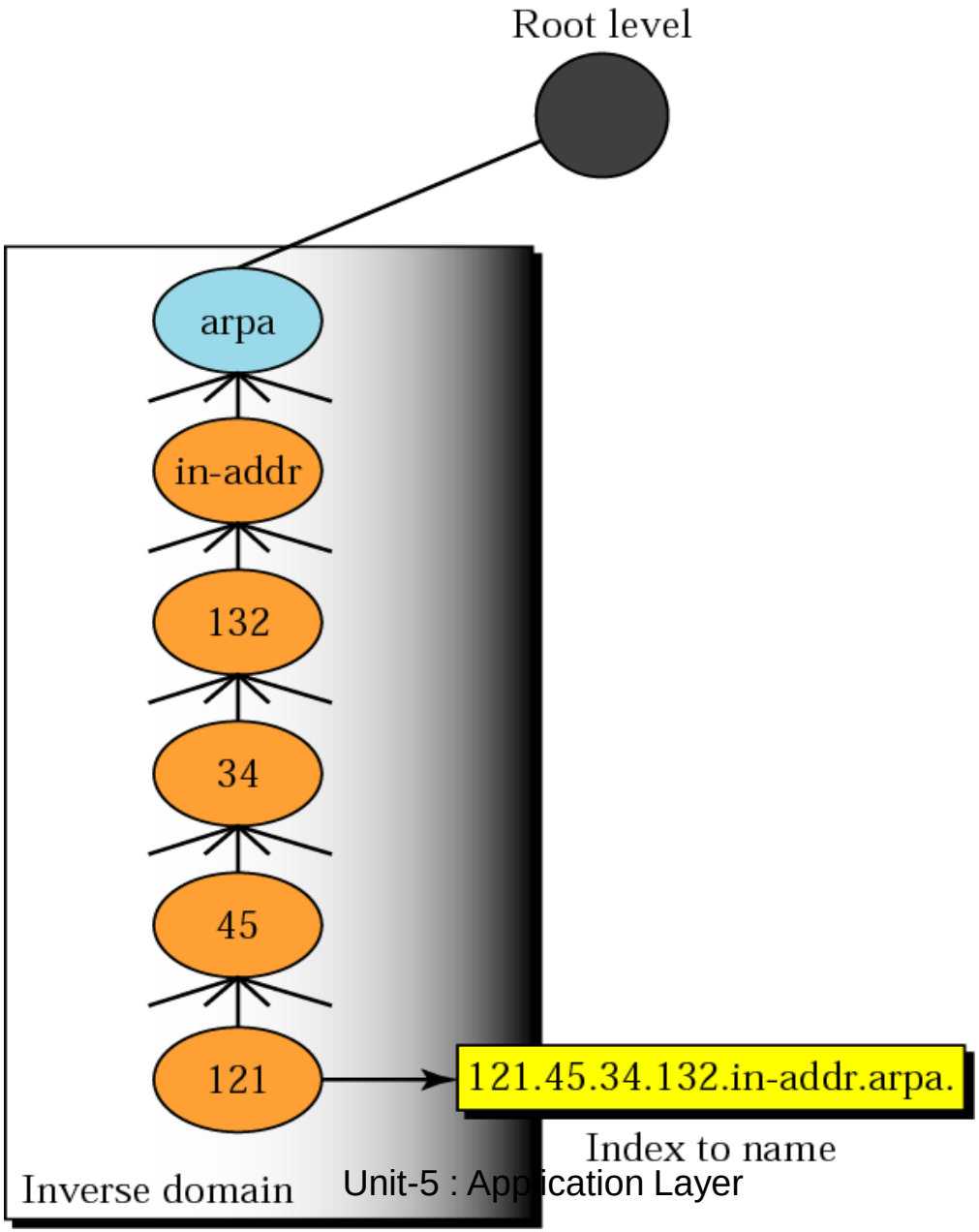


Figure 25.10 Inverse domain



25.5 Resolution

Resolver

Mapping Names to Addresses

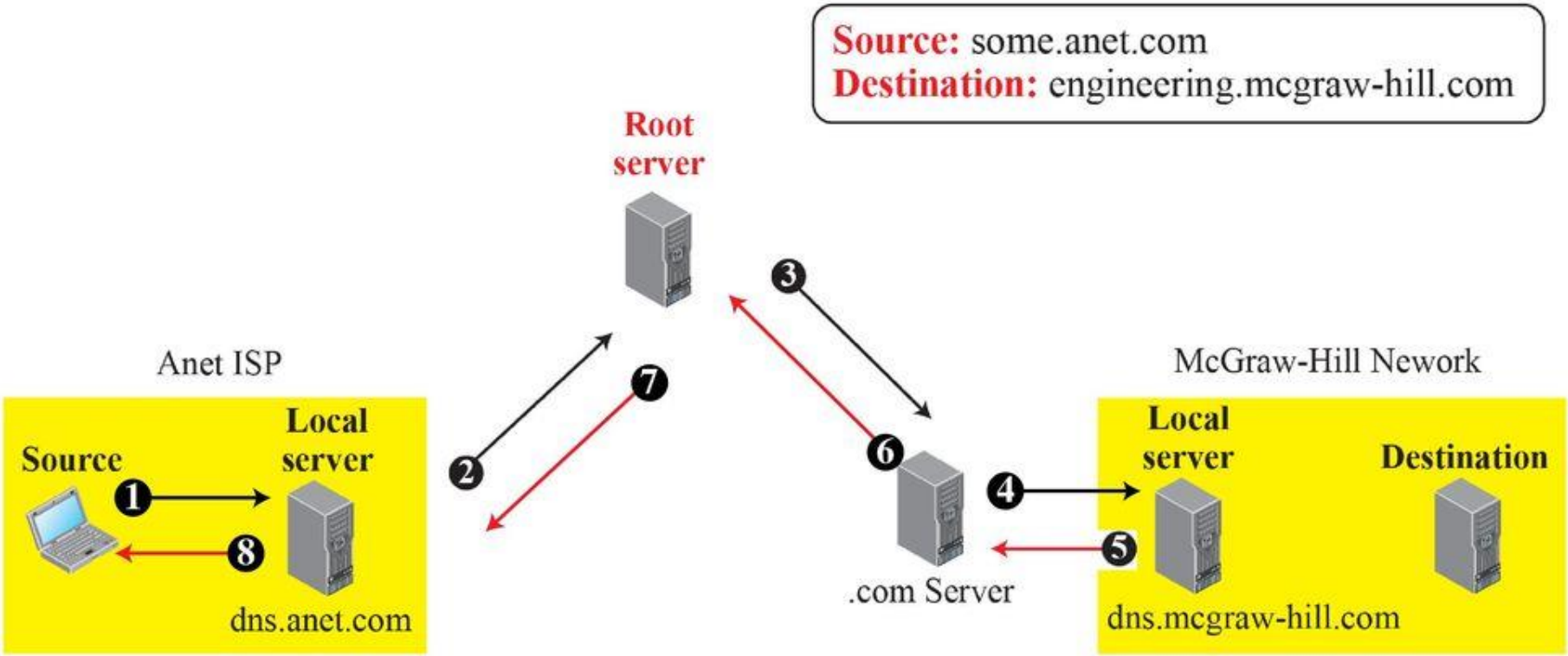
Mapping Addresses to Names

Recursive Resolution

Iterative Resolution

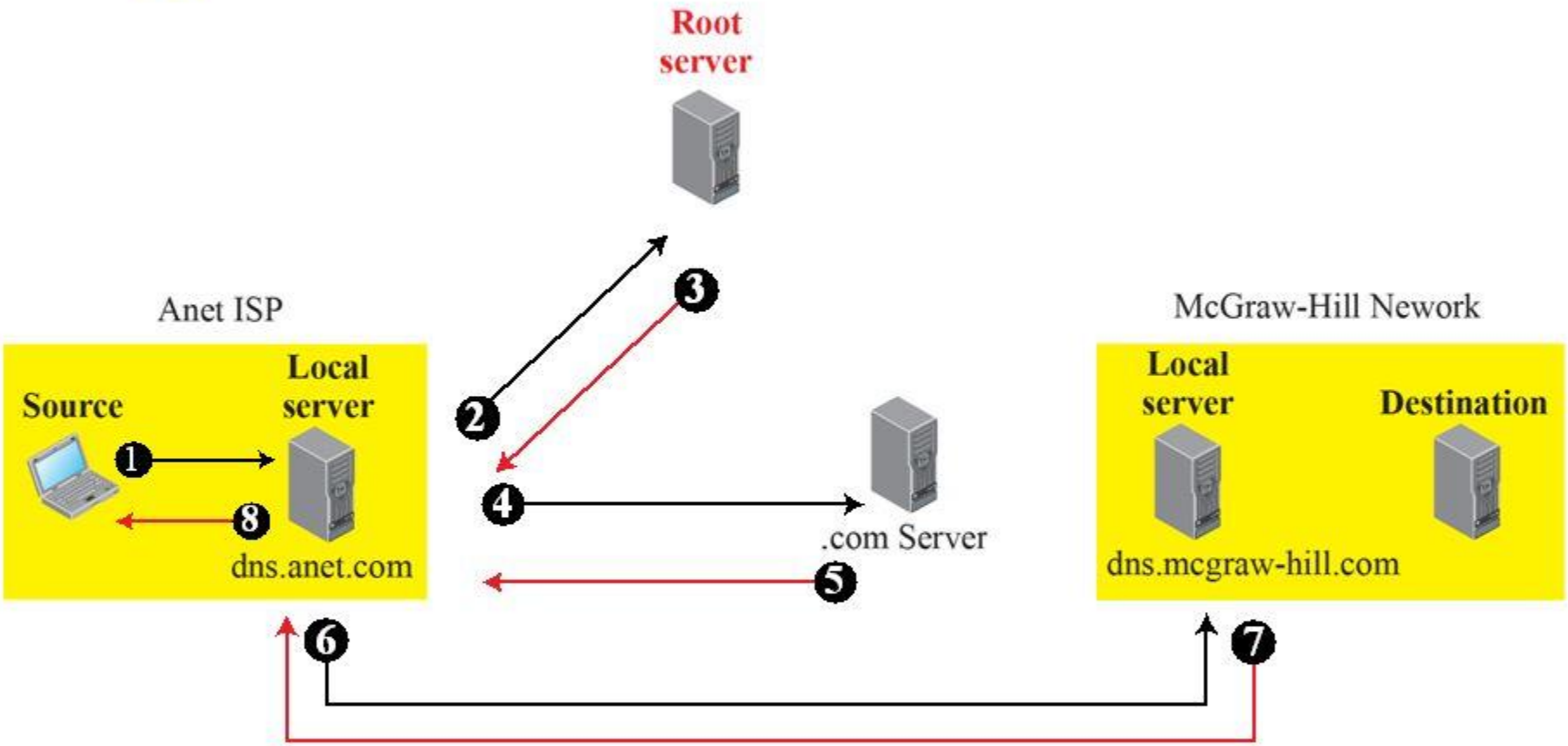
Caching

Figure 26.36: Recursive resolution



The Figure shows a simple example of a **recursive resolution.**
We assume that an application program running on a host named ***some.anet.com*** needs to find the IP address of another host named ***engineering.mcgraw-hill.com***.

Figure 26.37: Iterative resolution



Source: some.anet.com
Destination: engineering.mcgraw-hill.com

Resource Records

- Zone information -Set of resource records
- Resource record – 5 tuple

(Domain Name, Type, Class, TTL, Value)

25.6 DNS Messages

Header

Question Section

Answer Section

Authoritative Section

Additional Information Section

Figure 25.13 Query and response messages

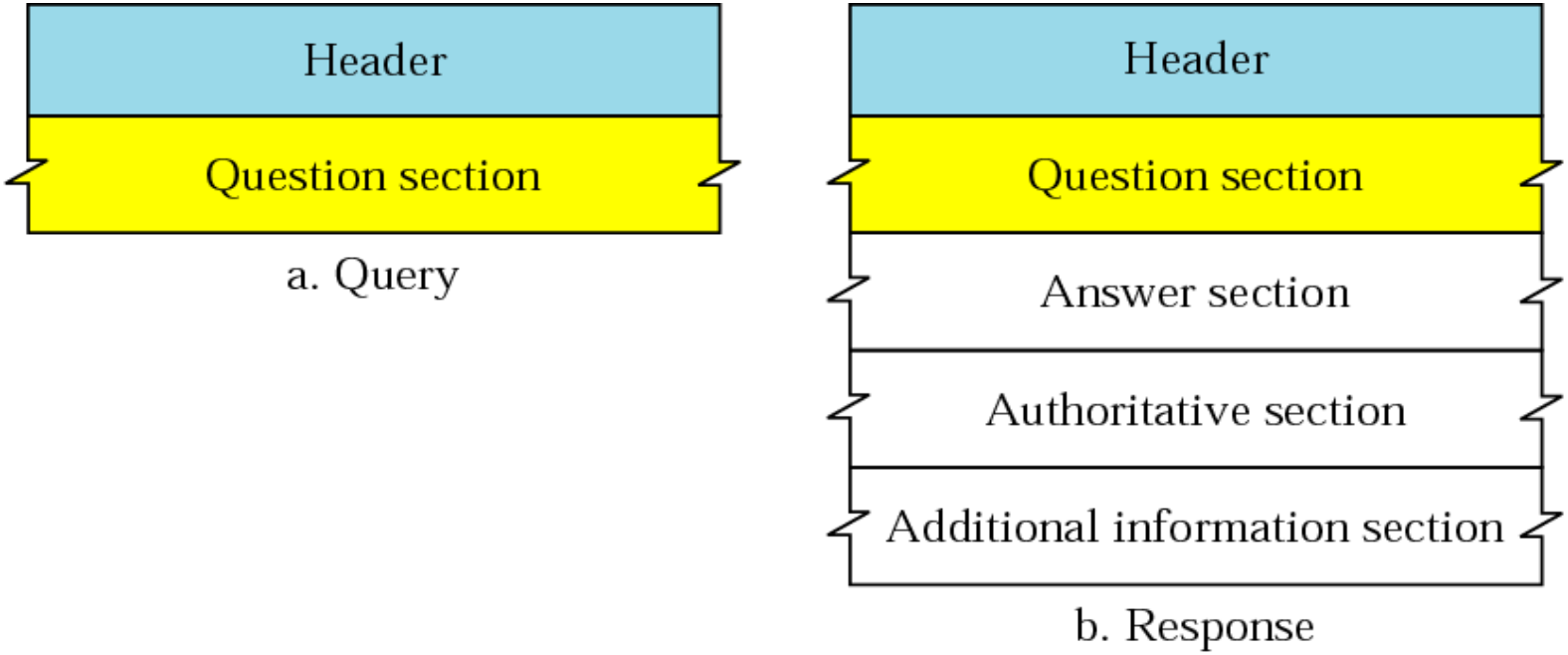


Figure 25.14 Header format

2 bytes	2 bytes
Identification	Flags
Number of question records	Number of answer records (All 0s in query message)
Number of authoritative records (All 0s in query message)	Number of additional records (All 0s in query message)

Registrars

- New domains added to DNS
- Accredited by ICANN
- Verifies whether it is unique
- Fee is charged
- Can find addresses at

  http://www.intenic.net
- To register
- Domain name: ws.wonderful.com
- IP address : 200.200.200.5

DDNS

- Sends update to primary DNS through DHCP
- Secondary servers are notified – actively or passively
- Active – primary sends message to secondary
- Passive – secondary periodically checks for changes

Security of DNS

- DNS messages are confidential to read profile of DNS servers
- Create a bogus reply to client – message origin authentication or integrity
- Flood DNS server – DOS attack

DNS can use the services of UDP or TCP, using the well-known port 53.

SMTP and FTP

26.1 Electronic Mail

Sending/Receiving Mail

Addresses

User Agent

MIME

Mail Transfer Agent

Mail Access Protocols

Figure 26.1 Format of an email

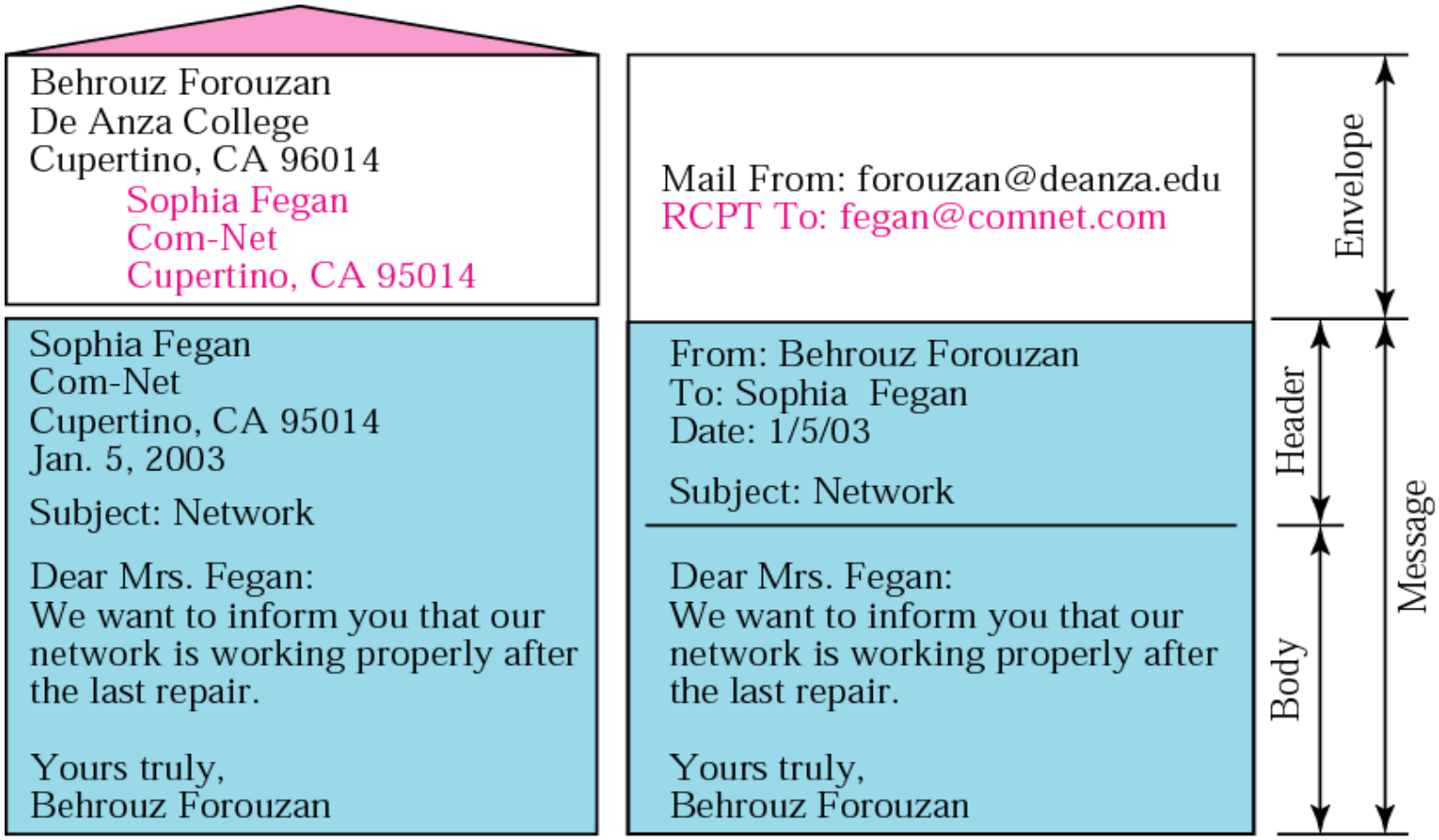


Figure 26.2 Email address

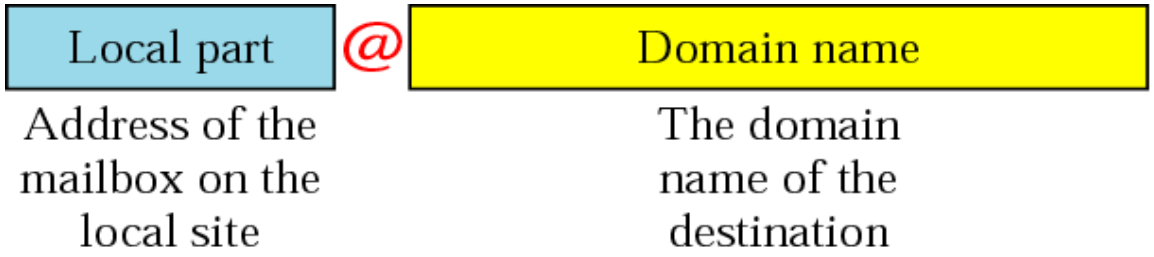
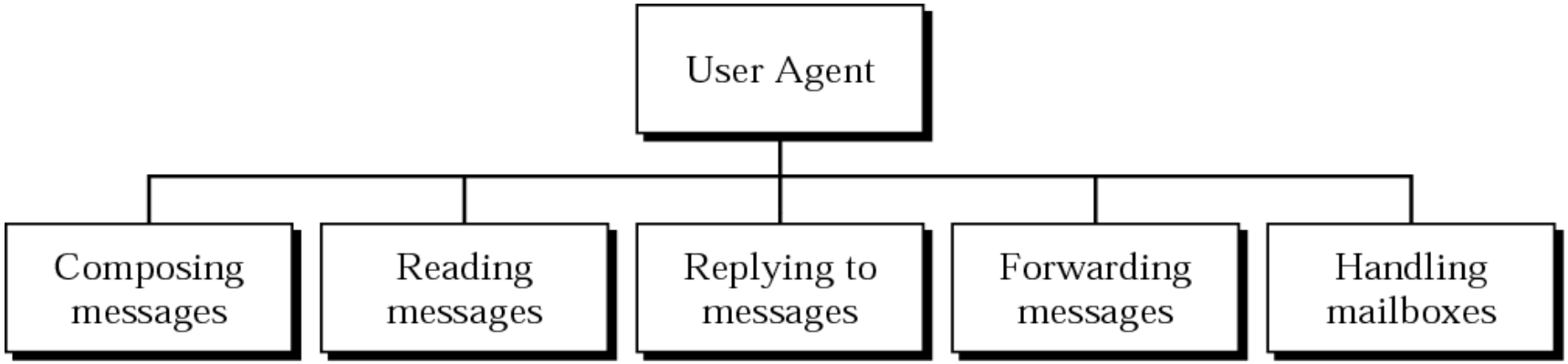


Figure 26.3 User agent



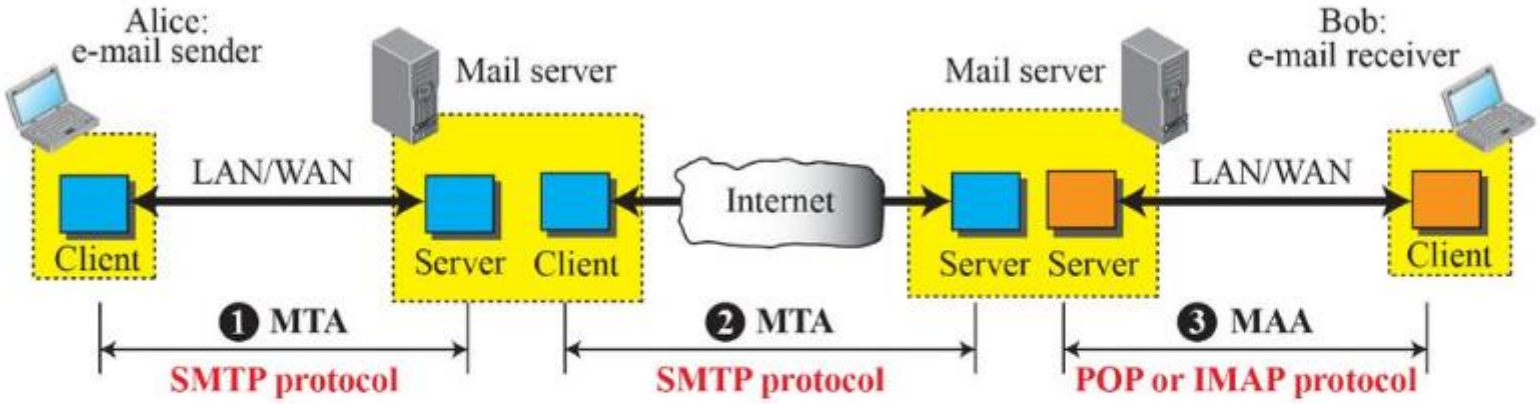
Some examples of command-driven user agents are mail, pine, and elm.

Some examples of GUI-based user agents are Eudora, Outlook, and Netscape.

Message Transfer Agent : SMTP

- It defines MTA client and server

Figure 26.15: ***Protocols used in electronic mail***



For MTA, the message needs to be pushed from the client to the server (need a *push* protocol) → ***Simple Mail Transfer Protocol (SMTP)***.

For MAA, the client must pull messages from the server (need a *pull* protocol) →***POP and IMAP.***

Commands

- Uses commands to transfer messages client and server

Table 26.6: SMTP Commands

The command is from an MTA client to an MTA server.

Keyword	*Argument(s)*	*Description*
HELO	Sender's host name	Identifies itself
MAIL FROM	Sender of the message	Identifies the sender of the message
RCPT TO	Intended recipient	Identifies the recipient of the message
DATA	Body of the mail	Sends the actual message
QUIT		Terminates the message
RSET		Aborts the current mail transaction
VRFY	Name of recipient	Verifies the address of the recipient
NOOP		Checks the status of the recipient
TURN		Switches the sender and the recipient
EXPN	Mailing list	Asks the recipient to expand the mailing list.
HELP	Command name	Asks the recipient to send information about the command sent as the argument
SEND FROM	Intended recipient	Specifies that the mail be delivered only to the terminal of the recipient, and not to the mailbox
SMOL FROM	Intended recipient	Specifies that the mail be delivered to the terminal *or* the mailbox of the recipient
SMAL FROM	Intended recipient	Specifies that the mail be delivered to the terminal *and* the mailbox of the recipient

Responses

- Sent from Server to Client

The response is from an MTA server to the MTA client

Code	Description
	Positive Completion Reply
211	System status or help reply
214	Help message
220	Service ready
221	Service closing transmission channel
250	Request command completed
251	User not local; the message will be forwarded
	Positive Intermediate Reply
354	Start mail input
	Transient Negative Completion Reply
421	Service not available
450	Mailbox not available
451	Command aborted: local error
452	Command aborted; insufficient storage
	Permanent Negative Completion Reply
500	Syntax error; unrecognized command
501	Syntax error in parameters or arguments
502	Command not implemented
503	Bad sequence of commands
504	Command temporarily not implemented
550	Command is not executed; mailbox unavailable
551	User not local
552	Requested action aborted; exceeded storage location
553	Requested action not taken; mailbox name not allowed
554	Transaction failed

Mail Transfer Phases

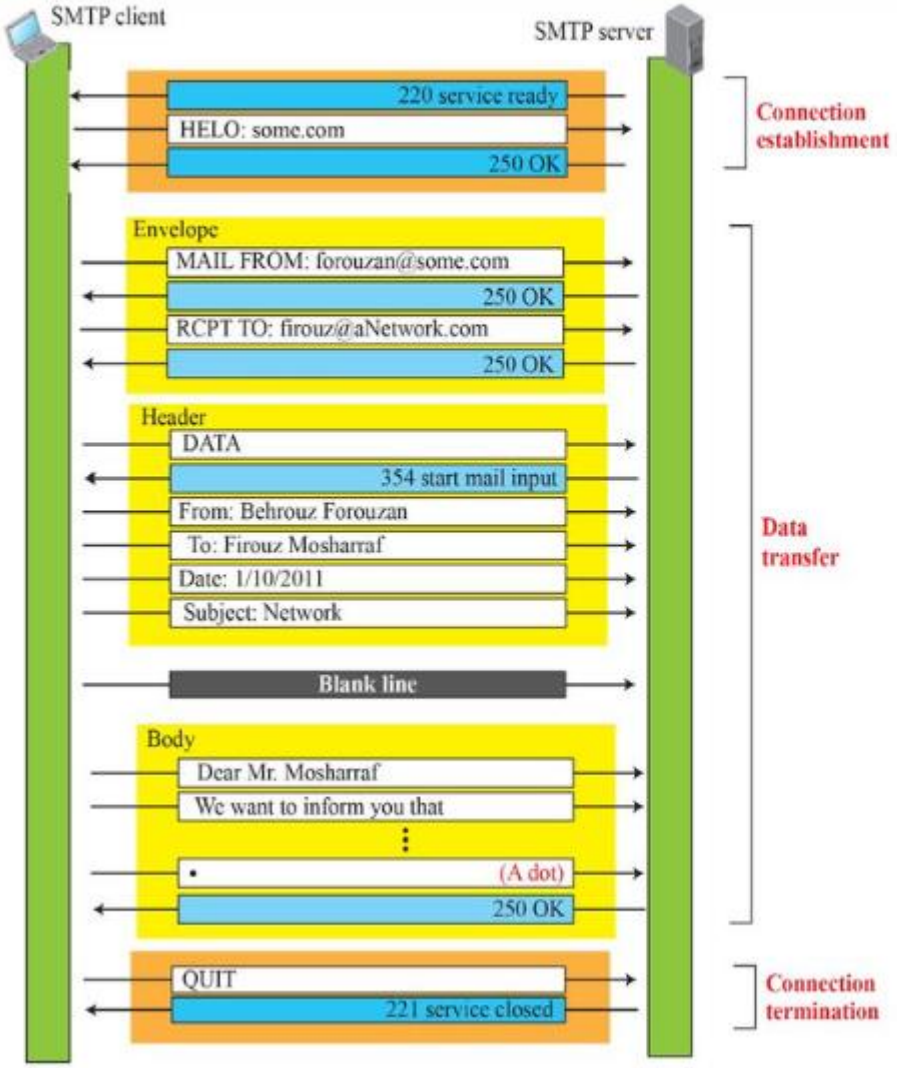
Example 26.12

The process of transferring a mail message occurs in three phases: connection establishment, mail transfer, and connection termination. In the figure, we have separated the messages related to the envelope, header, and body in the data transfer section.

Note that the steps in this figure are repeated two times in each e-mail transfer:

- once from the e-mail sender to the local mail server
- once from the local mail server to the remote mail server.

The local mail server, after receiving the whole e-mail message, may spool it and send it to the remote mail server at another time.



Message Access Agent: POP & IMAP

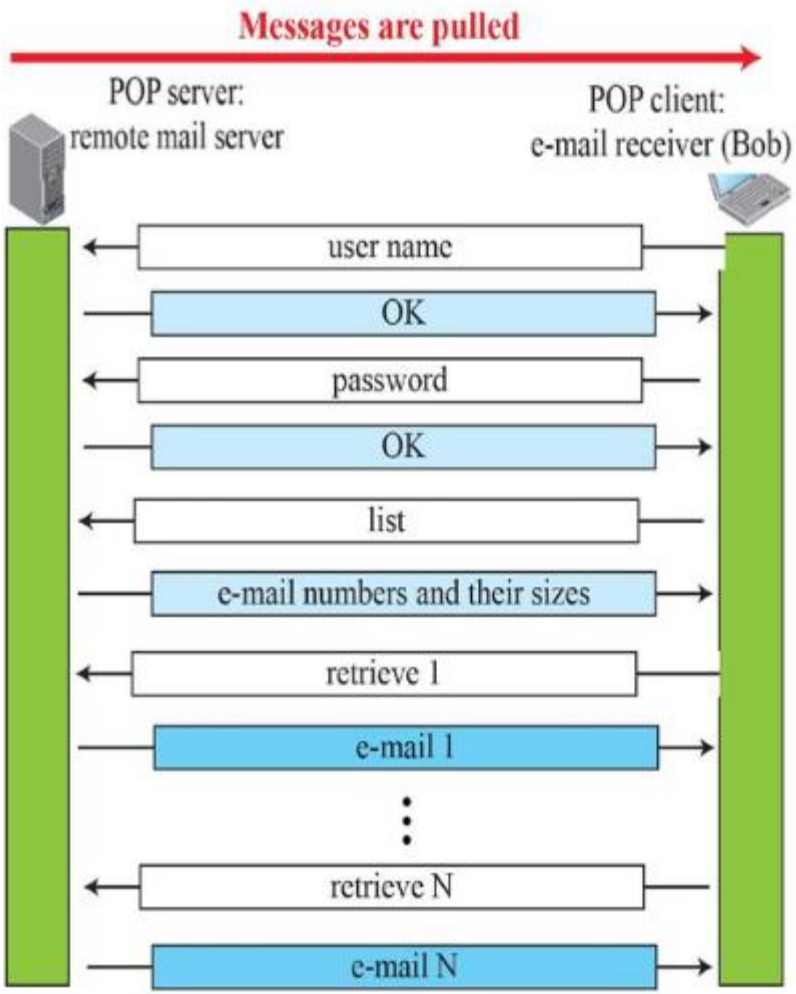
- Client pull a message from server – MAA
- Post Office Protocol – v3
- Internet Mail Access Protocol – IMAP4

POP3

Figure 26.17: POP3

Post Office Protocol, version 3 (POP3)

Mail access starts with the client when the user needs to download its e-mail from the mailbox on the mail server. The client opens a connection to the server on TCP port 110.
It then sends its user name and password to access the mailbox. The user can then list and retrieve the mail messages, one by one.



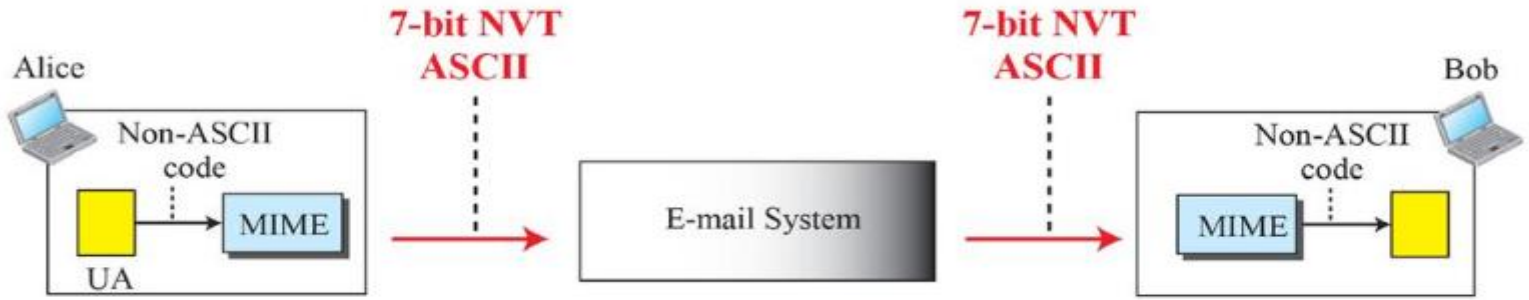
IMAP4

- User can email header prior to downloading
- Search contents of email
- Can partially download the email
- Create, delete, rename mailboxes on mail server
- Hierarchy of mail boxes

MIME

- Allow non ascii data to be sent through email

Figure 26.18: MIME



Electronic mail can send messages only in NVT 7-bit ASCII format. It cannot be used for languages other than English or to send binary files or video or audio data.

Multipurpose Internet Mail Extensions (MIME) is a supplementary protocol that transforms non-ASCII data at the sender site to NVT ASCII data and delivers it to the client MTA to be sent through the Internet. The message at the receiving site is transformed back to the original data.

Figure 26.5 MIME header

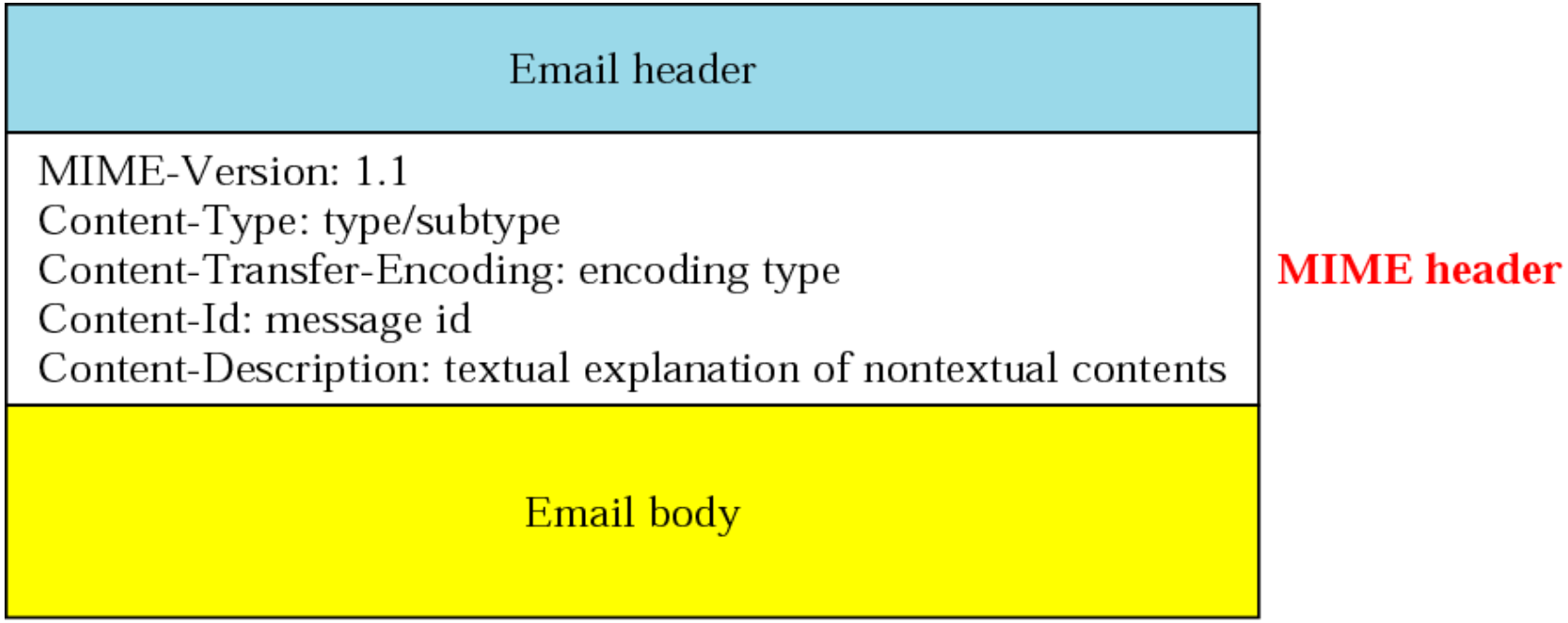


Table 26.1 *Data types and subtypes in MIME*

Type	Subtype	Description
Text	Plain	Unformatted text
Multiport	Mixed	Body contains ordered parts of different data types
	Parallel	Same as above, but no order
	Digest	Similar to mixed, but the default is message/RFC822
	Alternative	Parts are different versions of the same message
Message	RFC822	Body is an encapsulated message
	Partial	Body is a fragment of a bigger message
	Ext. Body	Body is a reference to another message
Image	JPEG	Image is in JPEG
	GIF	Video is in GIF format
Video	MPEG	Video is in MPEG format
Audio	Basic	Single-channel encoding of voice at 8 KHz
Application	PostScript	Adobe PostScript
	Octet-Stream	General binary data (8-bit bytes)

Table 26.2 Content-transfer encoding

Category	Description
Type	ASCII characters and short lines
7bit	Non-ASCII characters and short lines
8bit	Non-ASCII characters with unlimited-length lines
Binary	6-bit blocks of data are encoded into 8-bit ASCII characters
Base64	Non-ASCII characters are encoded as an equal sign followed by an ASCII code

Figure 26. 6 Base64

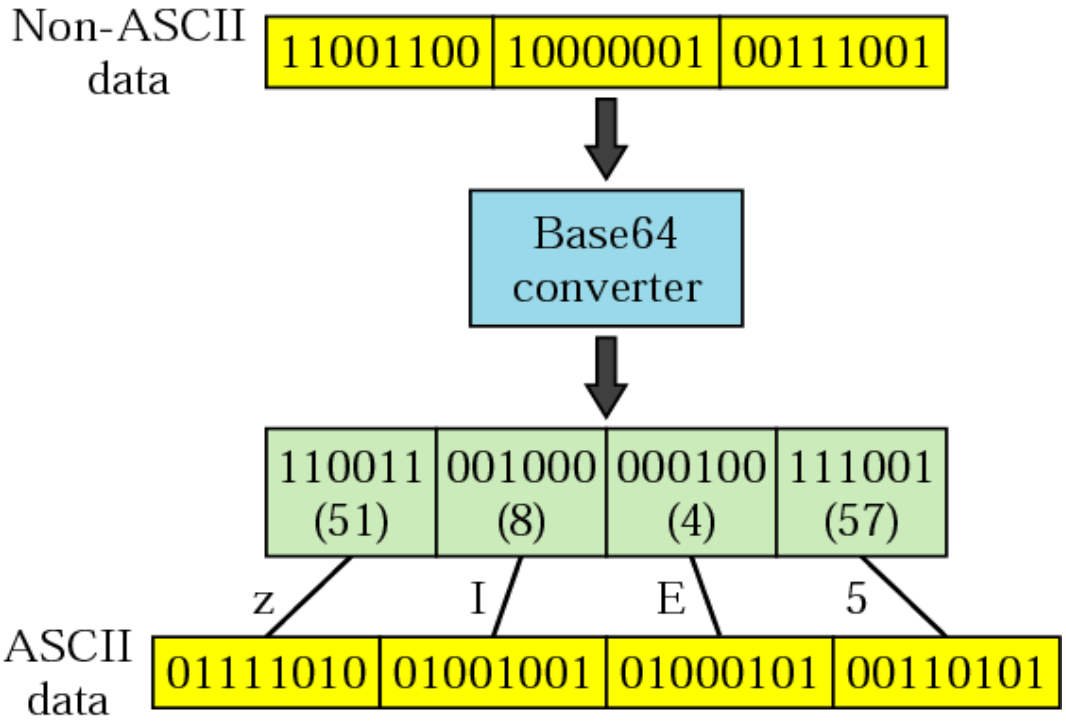
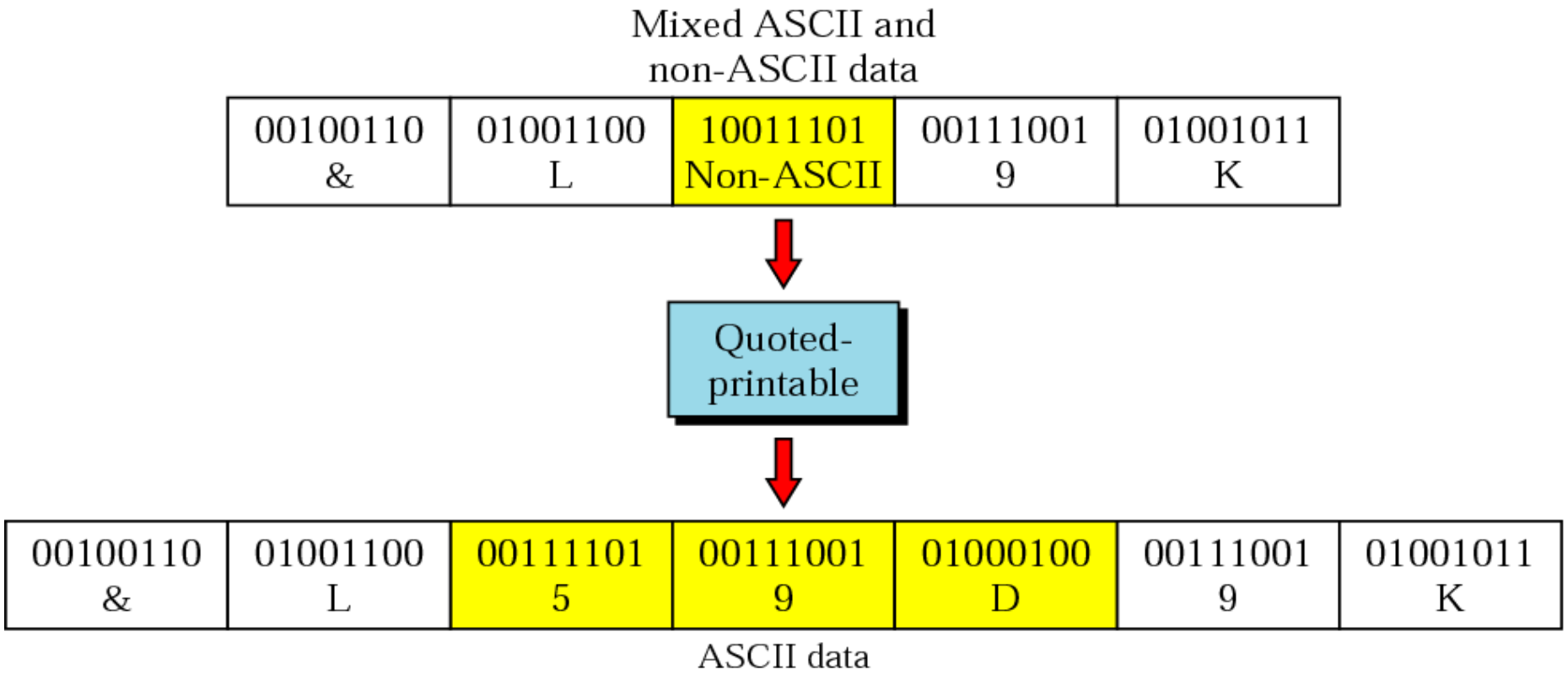


Table 26.3 Base64 encoding table

Value	Code	Value	Code	Value	Code	Value	Code	Value	Code	Value	Code
0	A	11	L	22	W	33	h	44	s	55	3
1	B	12	M	23	X	34	i	45	t	56	4
2	C	13	N	24	Y	35	j	46	u	57	5
3	D	14	O	25	Z	36	k	47	v	58	6
4	E	15	P	26	a	37	l	48	w	59	7
5	F	16	Q	27	b	38	m	49	x	60	8
6	G	17	R	28	c	39	n	50	y	61	9
7	H	18	S	29	d	40	o	51	z	62	+
8	I	19	T	30	e	41	p	52	0	63	/
9	J	20	U	31	f	42	q	53	1		
10	K	21	V	32	g	43	r	54	2		

Figure 26.7 Quoted-printable



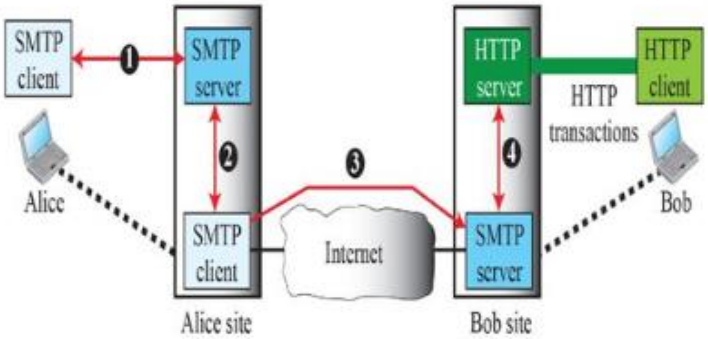
26.3.2 Web-Based Mail

E-mail is such a common application that some websites today provide this service to anyone who accesses the site.

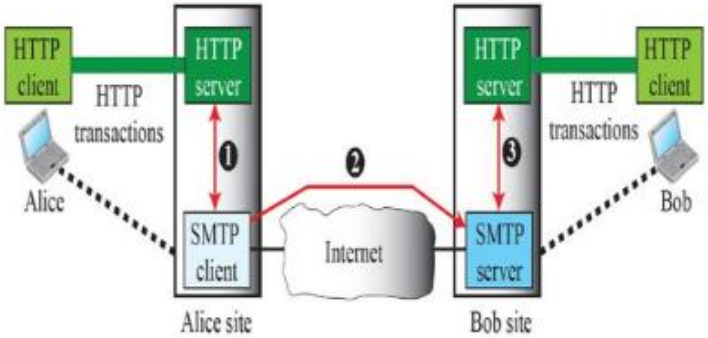
Three common sites are Hotmail, Yahoo, and Google mail. The idea is very simple. Figure 26.22 shows two cases.

Figure 26.22: Web-based e-mail, cases I and II

Case I: Alice, uses a mail server; Bob, the receiver, has an account on a web-based server.
Mail transfer from Alice's browser to her traditional mail serve, and from the sending mail server to the receiving mail server is through SMTP. However, the message from the receiving server (the web server) to Bob's browser is done through HTTP., instead of using POP3 or IMAP4.
When Bob needs to retrieve his e-mails, he sends a request HTTP message to the website (Hotmail, for example). The website sends a form to be filled in by Bob (the log-in name and the password).
If the log-in name and password match, the list of e-mails is transferred from the web server to Bob's browser in HTML format.
Case 2: In the second case, both Alice and Bob use web servers. Alice sends an HTTP request message to her web server using the name and address of Bob's mailbox as the URL. The server at the Alice site passes the message to the SMTP client and sends it to the server at the Bob site using SMTP protocol. Bob receives the message using HTTP transactions. SMTP protocol. Is used to transfer the message from the server at the Alice site to the server at the Bob site.



Case 1: Only receiver uses HTTP



Case 2: Both sender and receiver use HTTP

26.3.2 E-Mail Security

The protocol discussed in this chapter does not provide any security provisions per se. However, e-mail exchanges can be secured using two application-layer securities designed in particular for e-mail systems. Two of these protocols, Pretty Good Privacy (PGP) and Secure/Multipurpose Internet Mail Extensions (S/MIME), are discussed in Chapter 32 after we have discussed basic network security.

26.2 File Transfer

Connections

Communication

File Transfer

User Interface

Anonymous

FTP uses the services of TCP. It needs two TCP connections. The well-known port 21 is used for the control connection, and the well-known port 20 is used for the data connection.

Figure 26.12 FTP

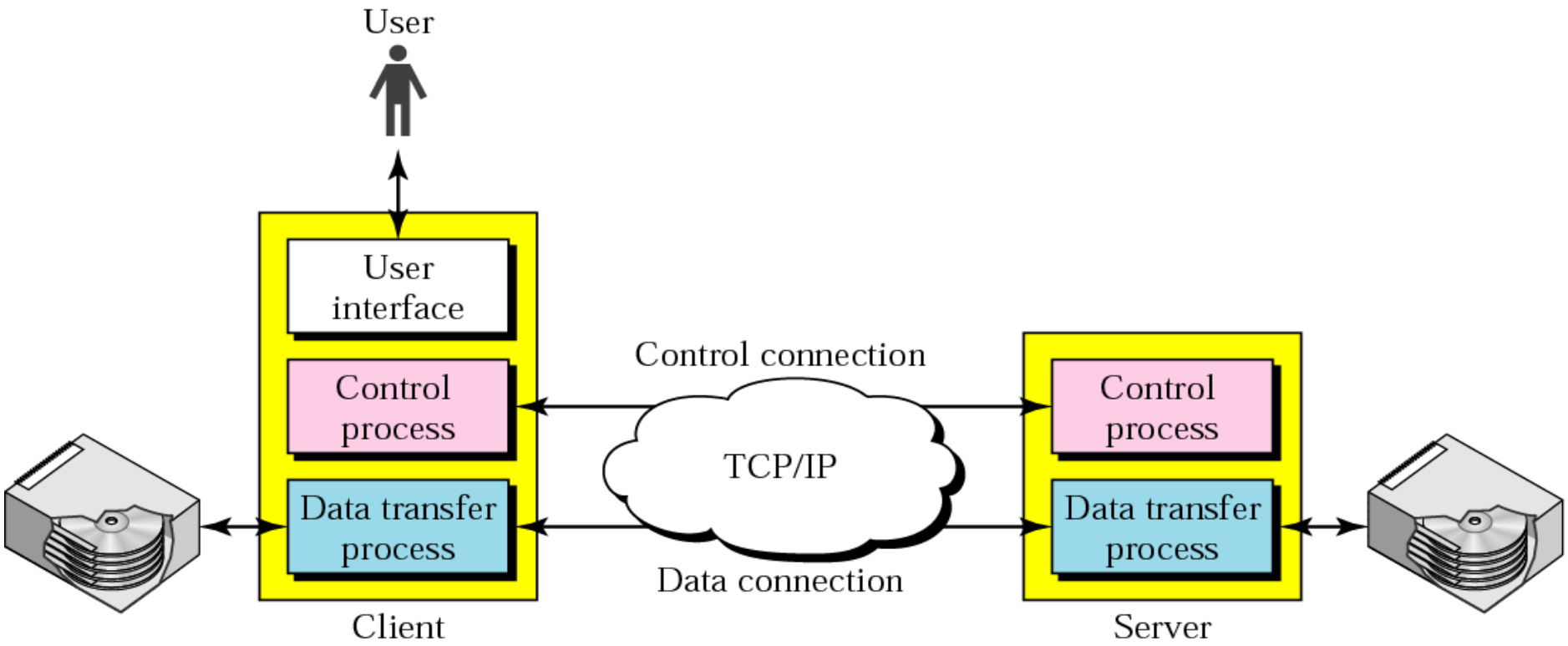


Figure 26.13 Using the control connection

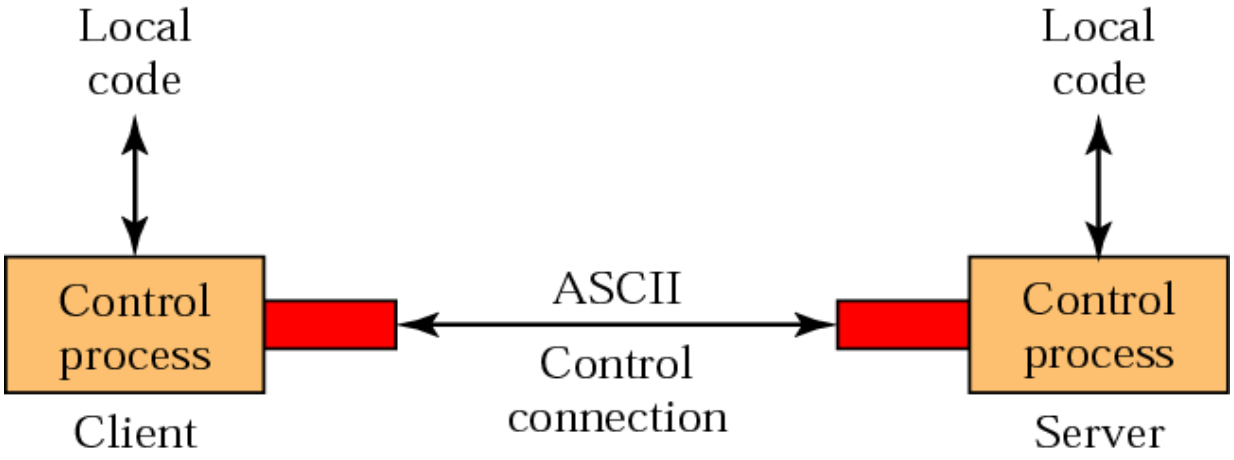


Figure 26.14 Using the data connection

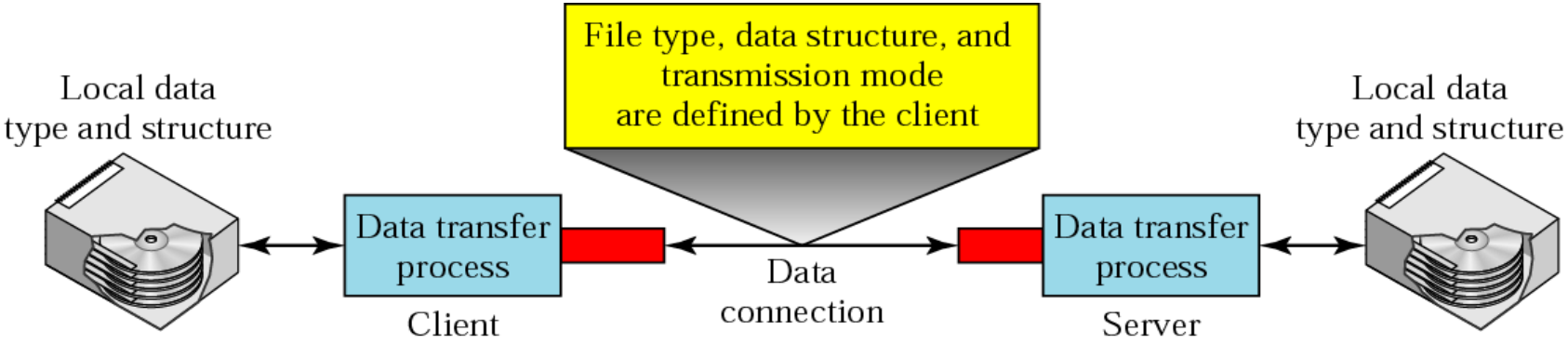
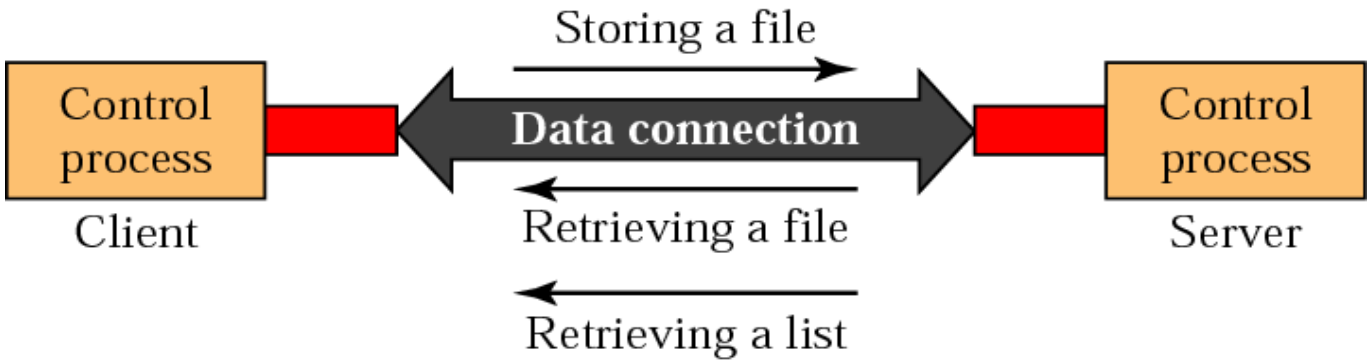


Figure 26.15 File transfer



Example 1

Figure 26.16 (next slide) shows an example of how a file is stored.

1. The control connection is created, and several control commands and responses are exchanged.
2. Data are transferred record by record.
3. A few commands and responses are exchanged to close the connection.

Figure 26.16 Example 1

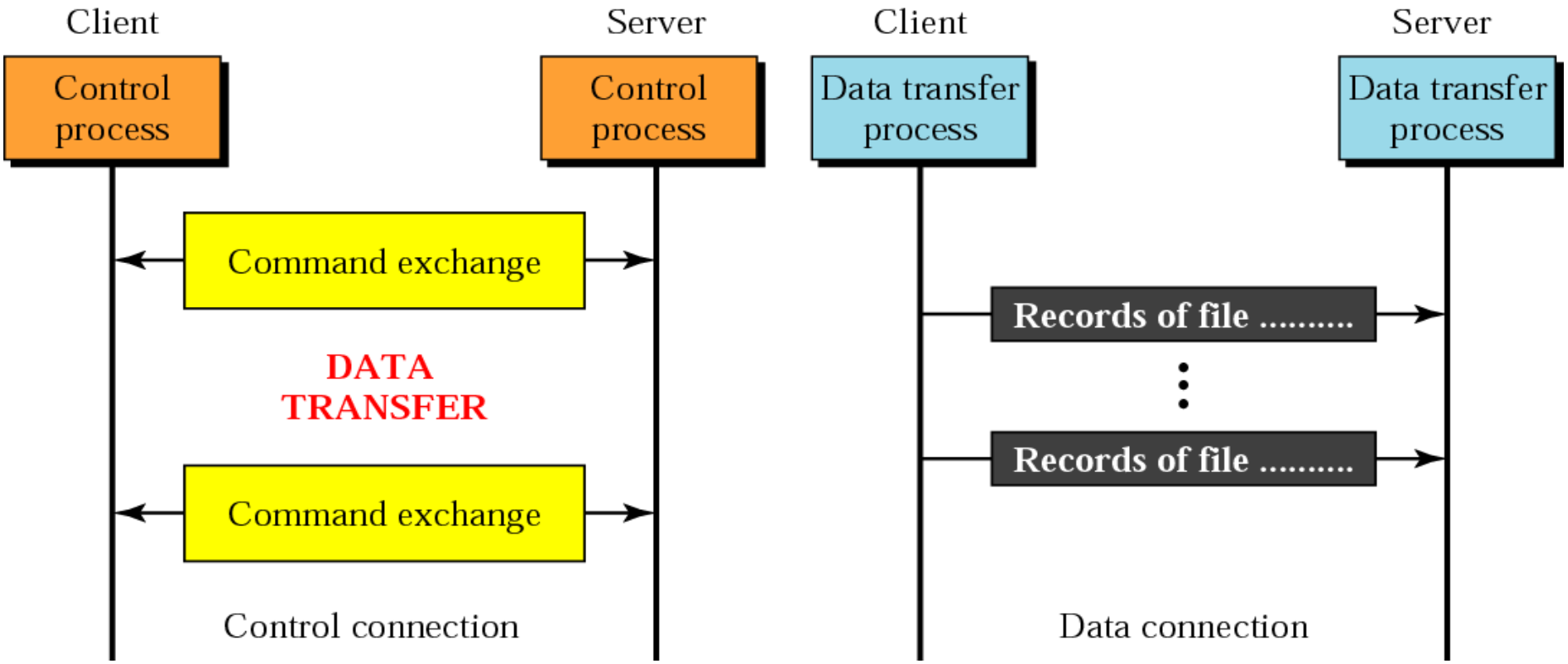


Table 26.4 List of FTP commands in UNIX

Commands
!, $, account, append, ascii, bell, binary, bye, case, cd, cdup, close, cr, delete, debug, dir, discount, form, get, glob, hash, help, lcd, ls, macdef, mdelete, mdir, mget, mkdir, mls, mode, mput, nmap, ntrans, open, prompt, proxy, sendport, put, pwd, quit, quote, recv, remotehelp, rename, reset, rmdir, runique, send, status, struct, sunique, tenex, trace, type, user, verbose,?

Example 2

We show some of the user interface commands that accomplish the same task as in Example 1. The user input is shown in boldface. As shown below, some of the commands are provided automatically by the interface. The user receives a prompt and provides only the arguments.

```
$ ftp challenger.atc.fhda.edu
Connected to challenger.atc.fhda.edu
220 Server ready
Name: forouzan
Password: xxxxxxx
ftp > ls /usr/user/report
200 OK
150 Opening ASCII mode
...........
...........
226 transfer complete
ftp > close
221 Goodbye
ftp > quit
```

Example 3

We show an example of using anonymous FTP. We connect to internic.net, where we assume there are some public data available.

```
$ ftp internic.net
Connected to internic.net
220 Server ready
Name: anonymous
331 Guest login OK, send "guest" as password
Password: guest
ftp > pwd
257 '/' is current directory
ftp > ls
200 OK
150 Opening ASCII mode
bin
...
ftp > close
221 Goodbye
ftp > quit
```

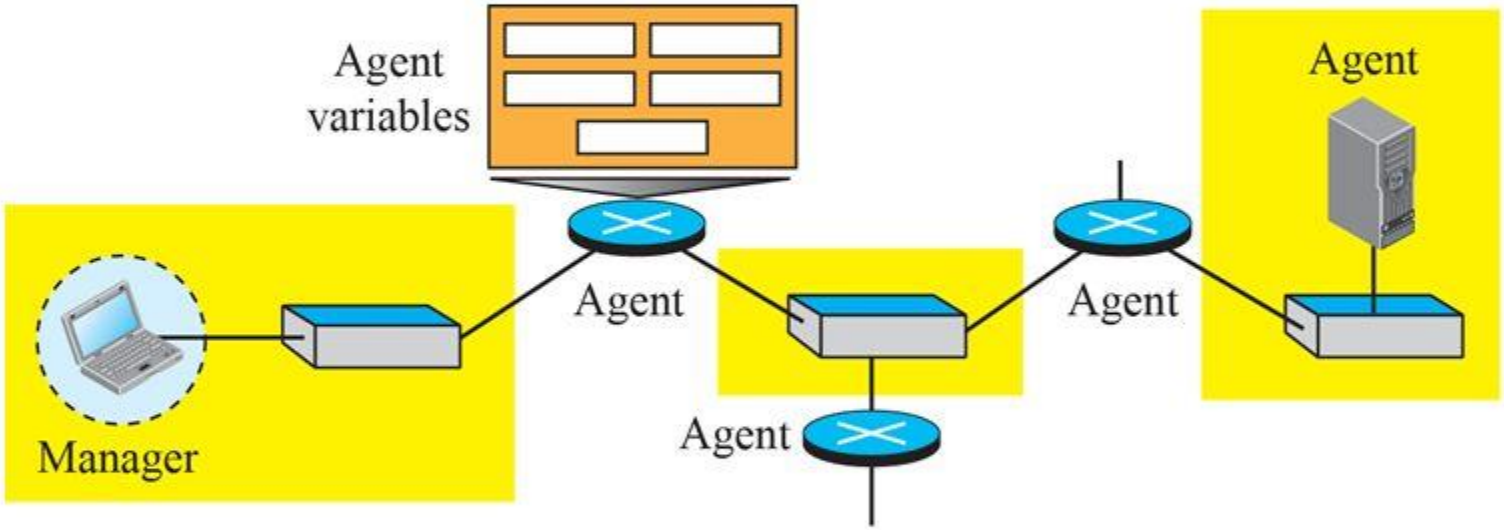
SNMP

What is SNMP?

- It is a set of several protocol definitions for managing networks.

- SNMP uses the concept of manager and agent , which means a manager controls and monitors a set of agents (usually routers).

Figure 27.2: SNMP concept



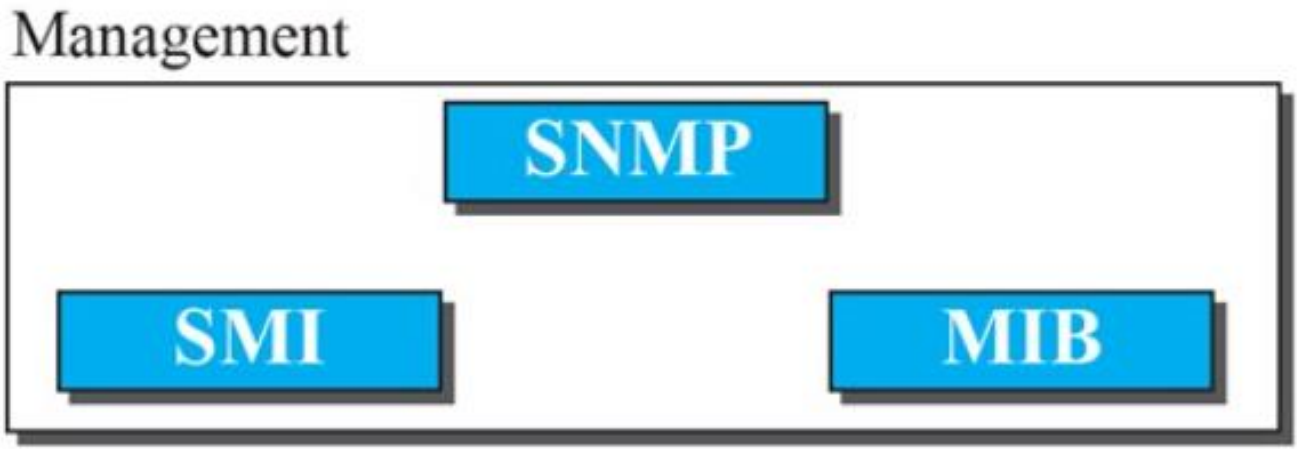
27.2.1 Managers and Agents

A management station, called a manager, is a host that runs the SNMP client program. A managed station, called an agent, is a router (or a host) that runs the SNMP server program. Management is achieved through simple interaction between a manager and an agent.

27.2.1 Managers and Agents

A management station, called a manager, is a host that runs the SNMP client program. A managed station, called an agent, is a router (or a host) that runs the SNMP server program. Management is achieved through simple interaction between a manager and an agent.

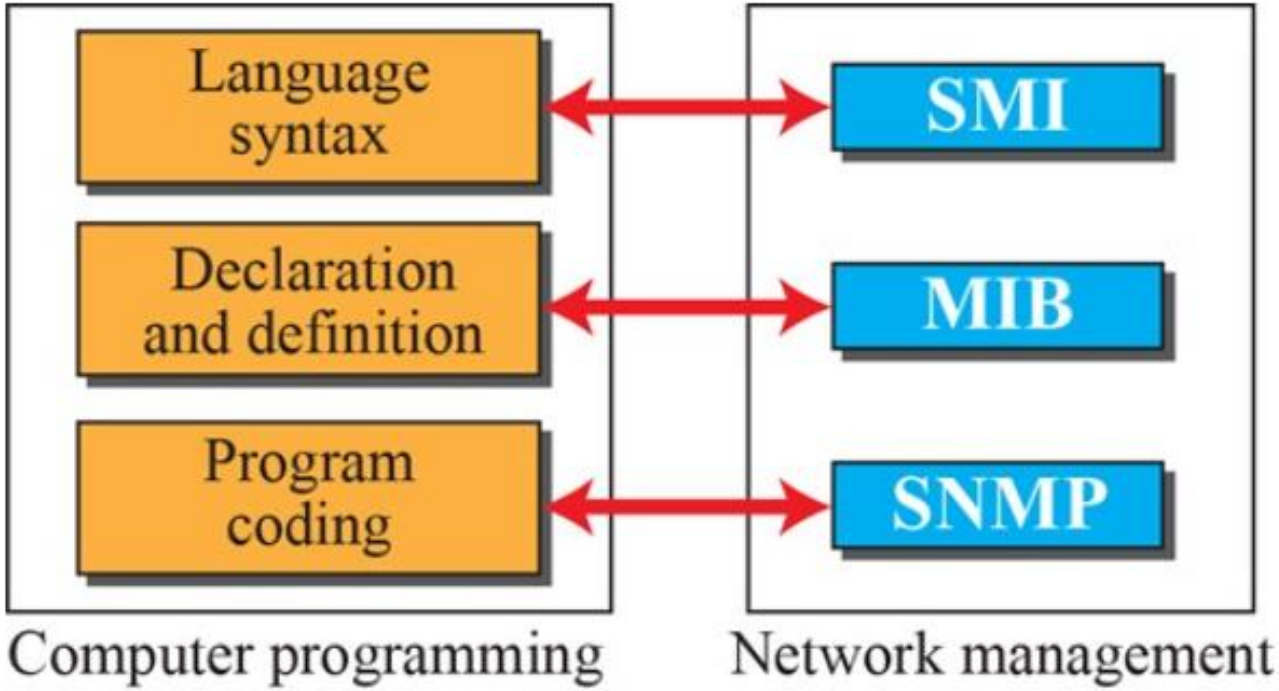
Figure 27.3: Components of network management on the Internet



ROLE

- SNMP – Formats of packet exchanged, reads and changes the status of objects being exchanged
- SMI – General rules for naming objects, defining object types, showing how to encode objects and value
- MIB – Creates a collection of named objects, their types and their relationships to each other in an entity to be managed

Figure 27.4: Comparing computer programming and network management



END

(Effort Never Dies)